In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [5]:
```
df=pd.read_excel('Mall_Customers - Mall_Customers.xlsx')
```

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype 
---  ------                  --------------  ----- 
 0   CustomerID              200 non-null    int64 
 1   Gender                  200 non-null    str   
 2   Age                     200 non-null    int64 
 3   Annual Income (k$)      200 non-null    int64 
 4   Spending Score (1-100)  200 non-null    int64 
dtypes: int64(4), str(1)
memory usage: 7.9 KB
```

In [7]:
```
df.head()
```

Out[7]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [33]:
```
g = sns.pairplot(df, height=1.7, aspect=1.2)
g.savefig("pairplot.pdf", bbox_inches="tight")
plt.show()
```



In [34]:
```
features = df.iloc[:, [3, 4]].values
```

In [17]:
```
from sklearn.cluster import KMeans

model = KMeans(n_clusters=5)
model.fit(features)
```

Out[17]:

▼ KMeans ⓘ ?

▶ Parameters

▶ Fitted attributes

▶ 5 features

In [21]:
```
model = KMeans(n_clusters=5, random_state=42, n_init=10)
model.fit(features)

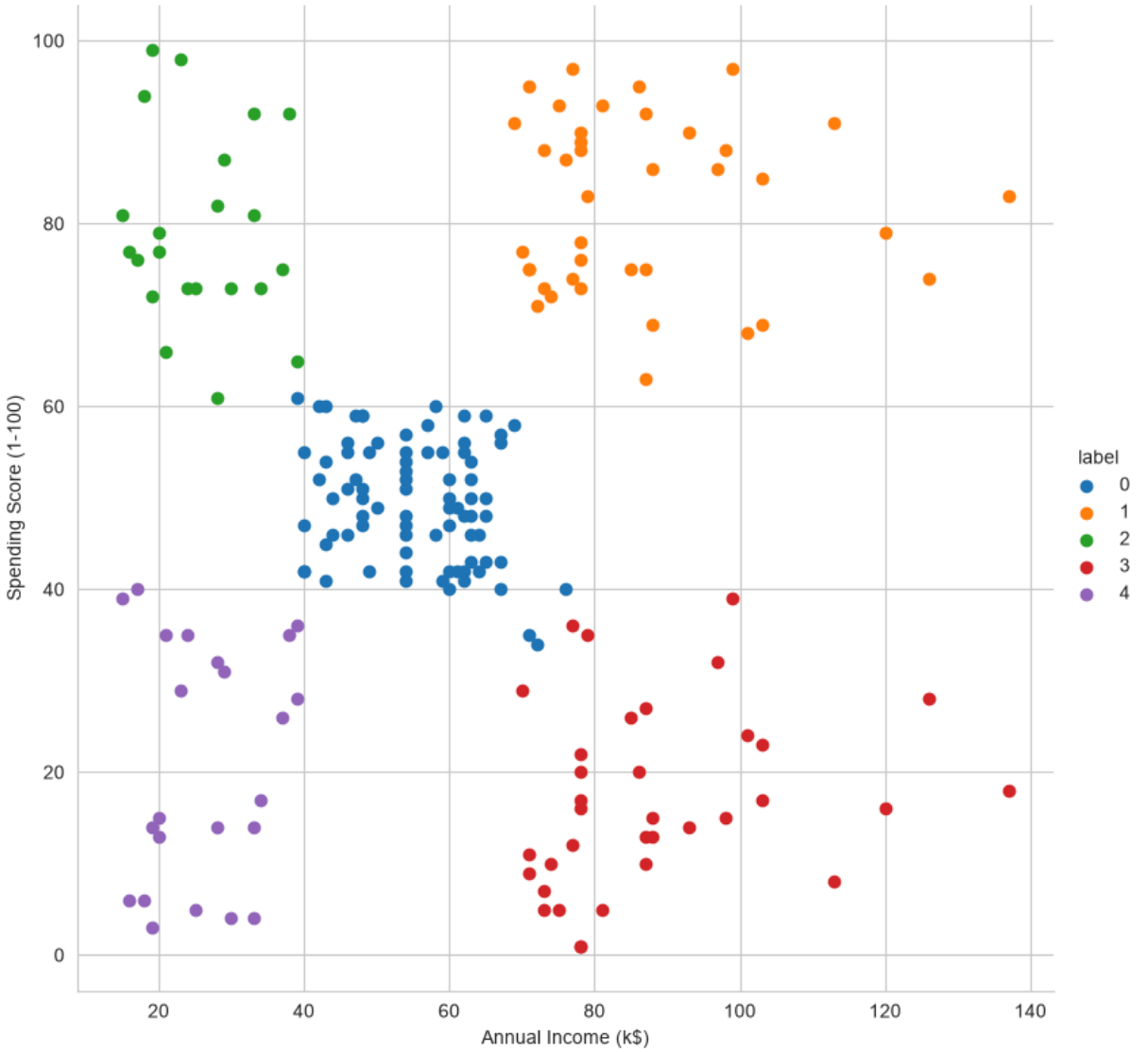
Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
Final.head()
```

Out[21]:

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

In [36]:

```
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
g.savefig("facetgrid.pdf", bbox_inches="tight")
plt.show()
```



In [25]:

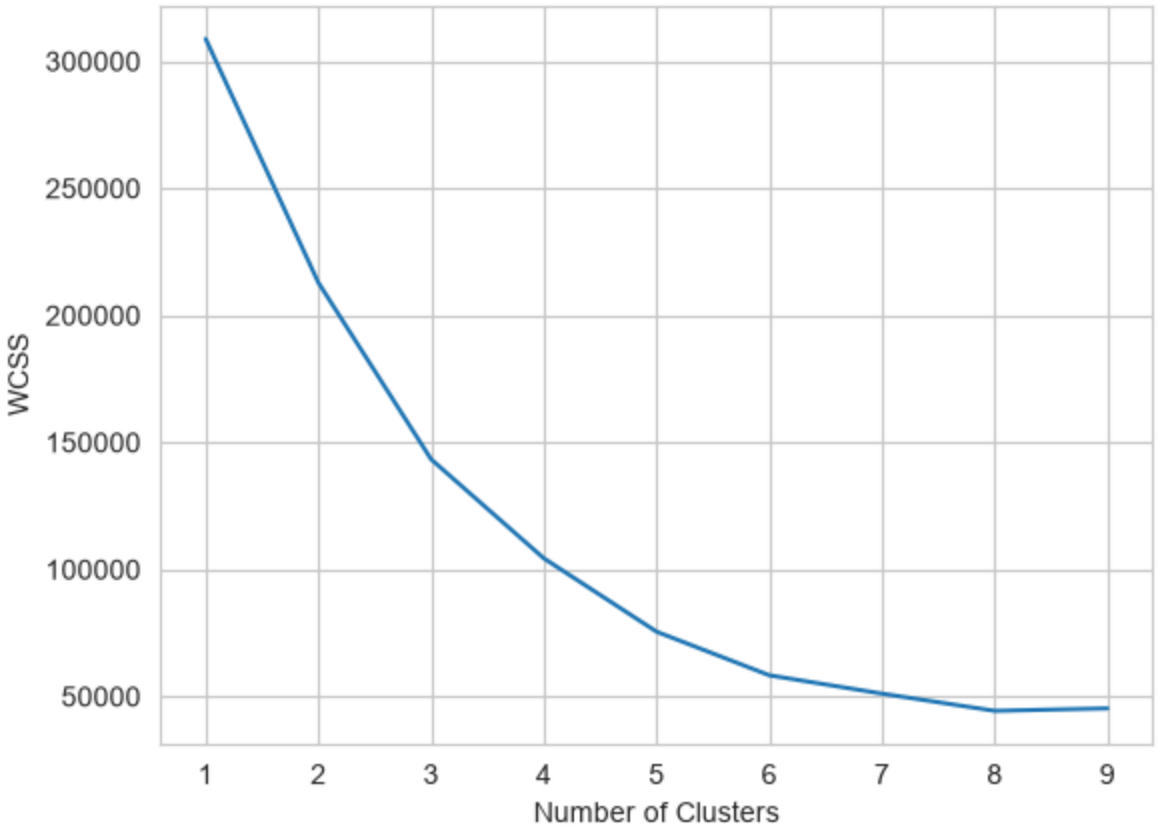
```
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

```
features_el = df.iloc[:, [2, 3, 4]].values

wcss = []

for i in range(1, 10):
    model = KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)

plt.plot(range(1, 10), wcss)
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```



In [ ]: